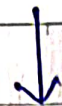


96

Paging vs Segmentation



Memory is divided into equal-sized parts of page

Memory is divided into variable-sized segments based on logical parts of the program.

ex. Imagine you have a word file with:

1. Title Page → 2 KB

2. Main Content → 5 KB

3. References → 3 KB

→ Assume that memory divided into blocks of 4 KB each.

① Paging (Fixed size Blocks)

• Page 1 (4 KB) → Part of Title Page + Part of Main Content

• Page 2 (4 KB) → Part of remaining main content (3 KB) + 1 KB of References

• Page 3 (4 KB) → Remaining 2 KB of ~~main content~~
↳ 2 KB empty

↳ Internal Fragmentation

② Segmentation (Variable Size blocks)

- Segment 1 (2 KB) → Title Page
- Segment 2 (5 KB) → Main Content
- Segment 3 (3 KB) → References

- External fragmentation occurs as processes are loaded & removed from the memory, the free memory space is broken into little pieces.

Page Fault : When a running process tries to access a page but the requested page is currently not loaded into main memory then it is called page fault.

→ When a page fault occurs, the OS has to replace an existing page with the newly required one.
- The process of replacement is sometimes called swap out or write to disk.

→ To ~~minim~~ minimize the number of page faults, ~~page~~ page replacement algorithms are used.
i.e. FIFO, LRU → Least Recently Used
↓
First In First Out

④① Page Replacement Algorithm - FIFO

FIFO - First In First Out

- This algorithm replaces the oldest page that has been present in the main memory for the longest time.
- whenever a page needs to be replaced, the first page in the queue is removed.

Advantages: It is simple and easy to use.

Disadvantages: The number of page faults increases by increasing the number of frames.

- Poor Performance.

Example: Consider page reference string
7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7
with 3 page frames.
Find the number of page faults & hits.

Reference String :

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7

		1	1	1	0	0	0	3	3		3	2		2
	0	0	0	3	3	3	2	2	2		1	1		1
7	7	7	2	2	2	4	4	4	0		0	0		7

Total number of page faults = 13

hits = 5

$$\text{Hit Ratio} = \frac{\text{No of hits}}{\text{no of references}} \times 100 = \frac{5}{18} \times 100 = 27.77\%$$

$$\text{Fault Ratio} = \frac{\text{No of faults}}{\text{no of ref. string}} \times 100 = \frac{13}{18} \times 100 = 72.22\%$$

1														
0		0												
1		1		1			0	4	1	1		2	3	1

Q2

Page Replacement Algo.: LRU

LRU - Least Recently Used

→ This algorithm replaces the page that has been used the least recently.

i.e. it replaces the page that has not been referenced for the longest time.

Advantages :

- more efficient than other replacement algorithms.

Disadvantages :

- expensive & complex to implement

Example : Consider page reference string 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7 with 3 page frames. find the no of page faults.

Reference string :

7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7

		1	1		3		3	2	2	2		2		2		7
	0	0	0		0		0	0	3	3		3		0		0
7	7	7	2		2		4	4	4	0		1		1		1

Page faults = 12

Page hits = 6

fault ratio = $\frac{12}{18} \times 100\% = \frac{2}{3} \times 100\% = 66.66\%$

page ratio = $\frac{6}{18} \times 100\% = \frac{1}{3} \times 100\% = 33.33\%$